

Mathematical modeling of an electrostatic MEMS with tilted elastomeric micro-pillars

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ABSTRACT

The aim of this study is to develop a comprehensive mathematical model for an electrostatic MEMS (Micro-Electro-Mechanical Systems) with gap-filling tilted micro-pillars. Elastomeric pillars are used due to their high dielectric constant, resulting in a higher equivalent permittivity of the gap medium. This leads to increased sensitivity and decreased required voltage. Although this system has gained popularity in recent years, a thorough understanding and accurate predictions of its behavior require a detailed mathematical model. Regrettably, the existing literature does not offer adequate information. The present model incorporates three coupled set of nonlinear differential equations that govern the longitudinal and transverse vibrations of the PDMS (Polydimethylsiloxane) pillars, as well as the transverse vibrations of the moving electrode. It demonstrates how the device's behavior is affected by varying tilt angles and numbers of micro-pillars. The viscoelastic properties of the PDMS pillars are also taken into account, following the Kelvin-Voigt model with non-linear strains. Numerical methods are used to solve the governing equations, enabling investigation of static deformation and vibrational responses under different DC (Direct Current) and DC+AC (Alternating Current) actuations. The nonlinear equations proposed in this study provide a valuable tool for accurately predicting the behavior of capacitive MEMS devices that use microstructured gaps with tilted micro-pillar arrays.

1. Introduction

Mathematical modeling plays a crucial role in comprehending the performance and behavior of various devices. It provides a framework for analyzing their complex dynamics and performance characteristics. By applying mathematical concepts, we can gain valuable insights, optimize device parameters, and predict the system's response to external stimuli. This paper presents a mathematical model and analysis of capacitive (electrostatic) microelectromechanical systems (MEMS) with microstructured elastomeric gaps of tilted micro-pillars that have gained interest in high-tech applications. The model addresses the interaction between electrostatic forces, mechanical deformation of the moving electrode, and viscoelastic deformation of the microstructured gap. A comprehensive mathematical description is required for accurate analysis of the coupled system that will be created.

MEMS is a technology that combines mechanical and electrical components on a microscopic scale. These systems have become

Abbreviations: MEMS, Micro-Electro-Mechanical Systems; DC, direct current; AC, Alternating current; SLM, Step-by-Step Linearization Method.

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increasingly popular due to their compatibility and ability to be utilized in various fields like aerospace, medicine, and telecommunications [1–6]. Some examples of MEMS applications can be seen in consumer electronics, energy harvesting, ultrasound transducers, loudspeakers, oscillators, scanning probe microscopes, automotive, industrial automation and manufacturing, defense, space, aeronautical, environmental and health sciences. MEMS technology is highly scalable in manufacturing, allowing for very low unit costs when produced in large quantities. They consume very little power and can easily be integrated with microelectronics to create embedded mechatronic systems. The possibilities for MEMS applications are virtually limitless, and new applications are being discovered regularly. Capacitive MEMS, a specific type of MEMS, shows great potential in sensing physical quantities due to its high sensitivity and accuracy [7–9].

In capacitive MEMS, the microstructured gap can be utilized for purposes such as increasing system sensitivity or reducing the required actuation voltage [10–12]. Polydimethylsiloxane (PDMS), Ecoflex, and polyurethane (PU) are among the various elastomeric soft materials that have a high dielectric constant (relative permittivity) and low Young's modulus, and have been widely used as the microstructured gap material [13–15], meanwhile, amorphous polymers such as PDMS are the most commonly used dielectric elastomers, due to their good elasticity and inherent flexibility, favorable biocompatibility and ease of fabrication. PDMS can be combined with other materials to expand its range of applications [14,15]. Previous studies have explored the use of bulk PDMS in the gap between capacitor electrodes, but its resistive compressive stiffness presents a significant mechanical challenge when used as a gap-filling material. To enhance the system's performance and sensitivity, one approach is to increase the deformability of the PDMS layer. This can be achieved by creating porosity in the layer [13,16,17]. Another way is to create a microstructured gap with a shaped PDMS layer, such as an array of micro-pillars with various geometries [18–20]. This technique can address the hysteresis of the PDMS viscoelastic material [20]. Among the various design options, the use of an array of tilted PDMS pillars has gained attention in MEMS technology [18]. Tilted pillars deform more easily under electrode pressure than vertical pillars, as they bend instead of compressing axially. This makes it easier to adjust the gap between the electrodes. Additionally, using tilted pillars reduces the air volume fraction caused by electrode movement, resulting in an increase in capacitance change. The deformation of tilted pillars is also amplified by the higher equivalent permittivity of the gap medium.

Elastomers, such as PDMS, have the ability to recover their original shape despite large deformations. Various constitutive models have been developed to describe the mechanical response of elastomers in different applications. One commonly used model is the linear elastic model, which assumes that the material deforms elastically up to a certain strain. Common models used to capture the highly nonlinear behavior of elastomers with large strains are the nonlinear hyperelastic models, such as the Mooney-Rivlin or Neo-Hookean models. These models characterize the material response by considering the strain energy density as a function of the strain tensor [21]. Viscoelastic models combine the elastic and viscous behavior of elastomers by considering their time-dependent response. Models such as the Maxwell or Kelvin-Voigt models contain parameters that represent a material's ability to dissipate energy and return to its original shape over time. The Kelvin-Voigt material, also known as a Voigt material, is the simplest model of a viscoelastic material that exhibits typical rubbery properties. It is perfectly elastic on long time scales (slow deformation), but more resistant to rapid deformation. The Kelvin-Voigt model consists of a fully viscous damper and a fully elastic spring connected in parallel. However, connecting these two elements in series results in a model of a Maxwell material [22].

There is an increasing interest in using an array of tilted micropillars as a microstructured gap in capacitive MEMS. Several empirical studies have been reported recently. To gain a thorough understanding and make accurate predictions of their behavior, a comprehensive mathematical model of such a system is required. However, current literature lacks sufficient information. This paper presents a mathematical model to demonstrate how the performance of a capacitive MEMS can be affected by using a microstructured gap with an array of tilted micro-pillars with varying tilt angles and numbers. The model includes three sets of differential equations governing the transverse vibrations of the moving electrode, and the longitudinal and bending vibrations of the PDMS pillars' array. The PDMS material conforms to the Kelvin-Voigt viscoelastic model, incorporating the non-linear strains. Numerical methods are used

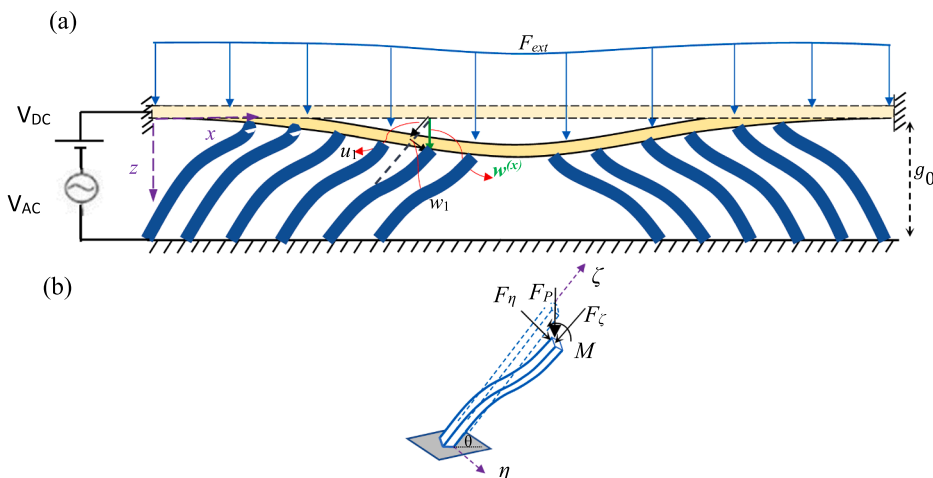


Fig. 1. a) A schematic view of a capacitive microbeam placed on an array of elastomeric tilted micro-pillars, and b) A deformed tilted micro-pillar.

to solve the governing equations. The static deformation and vibrational responses of the system's elements are addressed under common actuations (DC, step DC, and DC+AC voltages) based on the proposed equations.

2. Model description

Fig. 1(a) shows a schematic view of an electrostatically-actuated microbeam placed on an array of elastomeric micro-pillars. The micro-pillars are tilted, which causes their bending deformation. This makes it easier for the microbeam to deform compared to using vertical pillars. The bending deformation also increases the contribution of the elastomer in the gap volume between the top and bottom electrodes. The microbeam is double-clamped and homogeneous, and it is characterized by several parameters: Young's modulus (E), Poisson's ratio (ν), density (ρ), length (L), thickness (h), and width (b). It is supported by an array of micro-pillars with an angle θ . The subscript P denotes the elastomeric pillars. It is assumed that the pillars are strongly bonded to both the microbeam and bottom electrode. Upon application of an electrostatic field and/or an external force, the micro-beam deforms, causing the pillars to undergo longitudinal (u_1) and transversal deflection (w_1), with ξ and η as two orthogonal coordinates as shown in Fig. 1(b).

Before deriving the governing equations, a flowchart of the sequence of the main coupled equations is provided as follows (Fig. 2) for clarifying the complicated modeling process.

Now, we start deriving the required set of equations by introducing the Kelvin-Voigt linear viscoelastic constitutive model [22] for the pillar material as Eq. (1), where the nonlinear longitudinal strain is assumed for an element of the pillar.

$$\sigma_\xi = E_P \left(1 + \tau \frac{\partial}{\partial t} \right) \varepsilon_\xi = E_P \left(1 + \tau \frac{\partial}{\partial t} \right) \left(\frac{\partial u_1}{\partial \xi} - z \frac{\partial^2 w_1}{\partial \xi^2} + \frac{1}{2} \left(\frac{\partial w_1}{\partial \xi} \right)^2 \right) \quad (1)$$

The equation defines $E_P \tau$ as the viscosity of the elastomer and τ as the relaxation time [22]. Integrating the axial stress over the cross-sectional area of the pillar yields the axial force.

$$N = \int \int \sigma_\xi dA_P = E_P A_P \left(1 + \tau \frac{\partial}{\partial t} \right) \left(\frac{\partial u_1}{\partial \xi} + \frac{1}{2} \left(\frac{\partial w_1}{\partial \xi} \right)^2 \right) \quad (2)$$

where $A_P = b_P h_P$, and b_P and h_P are the width and thickness of the pillars, respectively. The equation for the longitudinal motion of a pillar's nonlinear vibration, taking into account nonlinear strains, is as follows:

$$E_P A_P \frac{\partial}{\partial \xi} \left[\left(1 + \tau \frac{\partial}{\partial t} \right) \left(\frac{\partial u_1}{\partial \xi} + \frac{1}{2} \left(\frac{\partial w_1}{\partial \xi} \right)^2 \right) \right] - \rho_P A_P \frac{\partial^2 u_1}{\partial t^2} = 0 \quad (3)$$

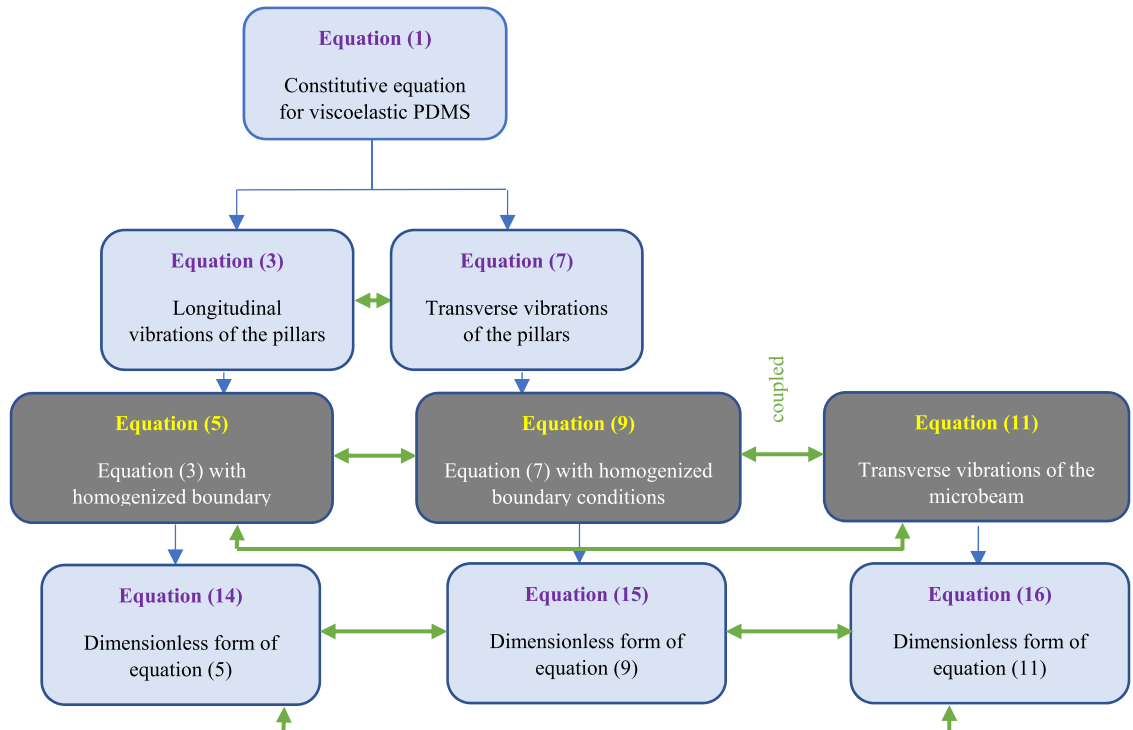


Fig. 2. A flowchart of the main equations' sequence.

with the following boundary conditions:

$$u_1(0, t) = 0 \text{ and } u_1^{(x)}(L_P, t) = -w^{(x)}(t)\sin\theta \quad (4)$$

To homogenize the boundary conditions, a new function for the longitudinal displacement can be defined as: $u_2(\xi, t) = u_1(\xi, t) + \frac{\xi}{L_P}w^{(x)}(t)\sin\theta$, where $w^{(x)}$ is the vertical displacement of the pillar's endpoint attached to the position “x” on the beam. This leads to the following equation and associated boundary conditions:

$$E_P A_P \left(\frac{\partial^2 u_2}{\partial \xi^2} + \frac{\partial w_2}{\partial \xi} \frac{\partial^2 w_2}{\partial \xi^2} + \tau \frac{\partial}{\partial t} \left(\frac{\partial^2 u_2}{\partial \xi^2} + \frac{\partial w_2}{\partial \xi} \frac{\partial^2 w_2}{\partial \xi^2} \right) \right) - \rho_P A_P \left(\frac{\partial^2 u_2}{\partial t^2} \right) = f_1^{(x)}(\xi, t)$$

where

$$\begin{aligned} f_1^{(x)}(\xi, t) = & -E_P A_P \left(6 \left(\frac{1}{L_P^2} - \frac{2\xi}{L_P^3} \right) \frac{\partial w_2}{\partial \xi} w^{(x)} \cos\theta + 6 \left(\frac{\xi}{L_P^2} - \frac{\xi^2}{L_P^3} \right) \frac{\partial^2 w_2}{\partial \xi^2} w^{(x)} \cos\theta + 36 \left(\frac{1}{L_P^2} - \frac{2\xi}{L_P^3} \right) \left(\frac{\xi}{L_P^2} - \frac{\xi^2}{L_P^3} \right) (w^{(x)})^2 \cos^2\theta + \tau \frac{\partial}{\partial t} \left(\right. \right. \\ & \left. \left. + 6 \left(\frac{1}{L_P^2} - \frac{2\xi}{L_P^3} \right) \frac{\partial w_2}{\partial \xi} w^{(x)} \cos\theta + 6 \left(\frac{\xi}{L_P^2} - \frac{\xi^2}{L_P^3} \right) \frac{\partial^2 w_2}{\partial \xi^2} w^{(x)} \cos\theta + 36 \left(\frac{1}{L_P^2} - \frac{2\xi}{L_P^3} \right) \left(\frac{\xi}{L_P^2} - \frac{\xi^2}{L_P^3} \right) (w^{(x)})^2 \cos^2\theta \right) \right) - \rho_P A_P \frac{\xi}{L_P} \frac{\partial^2 w^{(x)}}{\partial t^2} \sin\theta \end{aligned} \quad (5)$$

and

$$u_2(0, t) = 0 \text{ and } u_2^{(x)}(L_P, t) = 0 \quad (6)$$

The second equation of the coupled set describes the bending or transverse vibrations of the pillars. It takes into account the Kelvin-Voigt model with nonlinear strains and can be written as follows:

$$E_P I_P \left(1 + \tau \frac{\partial}{\partial t} \right) \frac{\partial^4 w_1}{\partial \xi^4} - E_P A_P \left(1 + \tau \frac{\partial}{\partial t} \right) \left[\frac{\partial^2 u_1}{\partial \xi^2} \frac{\partial w_1}{\partial \xi} + \frac{\partial u_1}{\partial \xi} \frac{\partial^2 w_1}{\partial \xi^2} + \frac{3}{2} \frac{\partial^2 w_1}{\partial \xi^2} \left(\frac{\partial w_1}{\partial \xi} \right)^2 \right] + \rho_P A_P \frac{\partial^2 w_1}{\partial t^2} = 0 \quad (7)$$

with the following boundary conditions:

$$w_1(0, t) = \frac{\partial w_1}{\partial \xi}(0, t) = 0, \quad w_1(L_P, t) = w^{(x)}(t)\cos\theta, \quad \frac{\partial w_1}{\partial \xi}(L_P, t) = 0 \quad (8)$$

Again, to homogenize the boundary conditions a new function is set as: $w_2(\xi, t) = w_1(\xi, t) - \xi^2 \left(\frac{3}{L_P^2} - \frac{2\xi}{L_P^3} \right) w^{(x)}(t)\cos\theta$. Therefore, Eq. (7) can be rewritten by substituting the new function as follows:

$$\begin{aligned} E_P I_P \frac{\partial^4 w_2}{\partial \xi^4} - E_P A_P \left(\frac{\partial^2 u_2}{\partial \xi^2} \frac{\partial w_2}{\partial \xi} + \frac{\partial u_2}{\partial \xi} \frac{\partial^2 w_2}{\partial \xi^2} \right) - \frac{3}{2} E_P A_P \frac{\partial^2 w_2}{\partial \xi^2} \left(\frac{\partial w_2}{\partial \xi} \right)^2 + E_P I_P \tau \frac{\partial}{\partial t} \frac{\partial^5 w_2}{\partial \xi^4} - E_P A_P \tau \frac{\partial}{\partial t} \left(\frac{\partial^2 u_2}{\partial \xi^2} \frac{\partial w_2}{\partial \xi} + \frac{\partial u_2}{\partial \xi} \frac{\partial^2 w_2}{\partial \xi^2} \right) \\ - \frac{3}{2} E_P A_P \tau \frac{\partial}{\partial t} \left(\frac{\partial^2 w_2}{\partial \xi^2} \left(\frac{\partial w_2}{\partial \xi} \right)^2 \right) + \rho_P A_P \frac{\partial^2 w_2}{\partial t^2} \\ = f_2^{(x)}(\xi, t) \end{aligned} \quad (9)$$

where

$$\begin{aligned} f_2^{(x)}(\xi, t) = & E_P A_P \left(6 \left(\frac{\xi}{L_P^2} - \frac{\xi^2}{L_P^3} \right) \frac{\partial^2 u_2}{\partial \xi^2} w^{(x)} \cos\theta + 6 \left(\frac{1}{L_P^2} - \frac{2\xi}{L_P^3} \right) \frac{\partial u_2}{\partial \xi} w^{(x)} \cos\theta - \frac{1}{L_P} \frac{\partial^2 w_2}{\partial \xi^2} w^{(x)} \sin\theta - 6 \frac{1}{L_P} \left(\frac{1}{L_P^2} - \frac{2\xi}{L_P^3} \right) w^{2(x)} \sin\theta \cos\theta \right) \\ & + \frac{3}{2} E_P A_P \left(12 \left(\frac{\xi}{L_P^2} - \frac{\xi^2}{L_P^3} \right) \frac{\partial w_2}{\partial \xi} \frac{\partial^2 w_2}{\partial \xi^2} w^{(x)} \cos\theta + 36 \left(\frac{\xi}{L_P^2} - \frac{\xi^2}{L_P^3} \right) \frac{\partial^2 w_2}{\partial \xi^2} w^{2(x)} \cos^2\theta + 6 \left(\frac{1}{L_P^2} - \frac{2\xi}{L_P^3} \right) \left(\frac{\partial w_2}{\partial \xi} \right)^2 w^{(x)} \cos\theta \right. \\ & \left. + 72 \left(\frac{\xi}{L_P^2} - \frac{\xi^2}{L_P^3} \right) \left(\frac{1}{L_P^2} - \frac{2\xi}{L_P^3} \right) \frac{\partial w_2}{\partial \xi} w^{2(x)} \cos^2\theta + 216 \left(\frac{\xi}{L_P^2} - \frac{\xi^2}{L_P^3} \right) \left(\frac{1}{L_P^2} - \frac{2\xi}{L_P^3} \right) w^{3(x)} \cos^3\theta \right) \\ & + E_P A_P \tau \frac{\partial}{\partial t} \left(6 \left(\frac{\xi}{L_P^2} - \frac{\xi^2}{L_P^3} \right) \frac{\partial^2 u_2}{\partial \xi^2} w^{(x)} \cos\theta + 6 \left(\frac{1}{L_P^2} - \frac{2\xi}{L_P^3} \right) \frac{\partial u_2}{\partial \xi} w^{(x)} \cos\theta - \frac{1}{L_P} \frac{\partial^2 w_2}{\partial \xi^2} w^{(x)} \sin\theta - 6 \frac{1}{L_P} \left(\frac{1}{L_P^2} - \frac{2\xi}{L_P^3} \right) (w^{(x)})^2 \sin\theta \cos\theta \right) \\ & + \frac{3}{2} E_P A_P \tau \left(12 \left(\frac{\xi}{L_P^2} - \frac{\xi^2}{L_P^3} \right) \frac{\partial w_2}{\partial \xi} \frac{\partial^2 w_2}{\partial \xi^2} w^{(x)} \cos\theta + 36 \left(\frac{\xi}{L_P^2} - \frac{\xi^2}{L_P^3} \right) \frac{\partial^2 w_2}{\partial \xi^2} (w^{(x)})^2 \cos^2\theta \right. \\ & \left. + 6 \left(\frac{1}{L_P^2} - \frac{2\xi}{L_P^3} \right) \left(\frac{\partial w_2}{\partial \xi} \right)^2 w^{(x)} \cos\theta + 72 \left(\frac{\xi}{L_P^2} - \frac{\xi^2}{L_P^3} \right) \left(\frac{1}{L_P^2} - \frac{2\xi}{L_P^3} \right) \frac{\partial w_2}{\partial \xi} (w^{(x)})^2 \cos^2\theta + 216 \left(\frac{\xi}{L_P^2} - \frac{\xi^2}{L_P^3} \right) \left(\frac{1}{L_P^2} - \frac{2\xi}{L_P^3} \right) (w^{(x)})^3 \cos^3\theta \right) \end{aligned}$$

$$-\rho_p A_p \left(\frac{3\xi^2}{L_p^2} - \frac{2\xi^3}{L_p^3} \right) \frac{\partial^2 w^{(x)}}{\partial t^2} \cos\theta$$

with the following boundary conditions:

$$w_2(0, t) = \frac{\partial w_2}{\partial \xi}(0, t) = 0, \quad w_2(L_p, t) = \frac{\partial w_2}{\partial \xi}(L_p, t) = 0 \quad (10)$$

Now, it is time to introduce the third equation which governs the transverse vibrations of the microbeam under the applied external forces as follows:

$$\left(\frac{EI}{1-\nu^2} \right) \frac{\partial^4 w}{\partial x^4} - \frac{EA}{2L} \left(\int_0^L \left(\frac{\partial w}{\partial x} \right)^2 dx \right) \frac{\partial^2 w}{\partial x^2} + \rho A \frac{\partial^2 w}{\partial t^2} + C \frac{\partial w}{\partial t} = f_3^{(x)}(t) + f_e(x, t) \quad (11)$$

where

$$f_3^{(x)} = n_p b \left(E_p A_p \left(1 + \tau \frac{\partial}{\partial t} \right) \left(\frac{\partial u_2}{\partial \xi} \Big|_{\xi=L_p} - \frac{1}{L_p} w^{(x)} \sin\theta \right) \sin\theta - E_p I_p \left(1 + \tau \frac{\partial}{\partial t} \right) \left(\frac{\partial^3 w_2}{\partial \xi^3} \Big|_{\xi=L_p} - \frac{12}{L_p^3} w^{(x)} \cos\theta \right) \cos\theta \right)$$

with the following boundary conditions:

$$w(0, t) = w(L, t) = 0 \text{ and } \frac{\partial w}{\partial x}(0, t) = \frac{\partial w}{\partial x}(L, t) = 0 \quad (12)$$

It should be noted that the total force exerted on the microbeam by resistive pillars, can be defined as the sum of the forces of each individual pillar, represented by the equation $\sum_{i=1}^{N_p} F_p^{(i)} \delta(x - x^{(i)})$. For a large number of pillars the force can be estimated using a continuous function by defining the pillars density as $n_p = \frac{N_p}{bL}$, where N_p is the number of pillars. $f_e(x, t)$ is the external force per unit length that can be a combination of electrostatic force per unit length $\left(\frac{\epsilon_{eff} \epsilon_0 b V^2}{2(g_0 - w)^2} \right)$ and any other external stimuli, where ϵ_{eff} and ϵ_0 are the dielectric constant of the gap medium and vacuum permittivity, respectively, and $g_0 = L_p \sin\theta$ is the initial gap between two parallel electrodes. V is the applied voltage and can generally be a DC bias voltage or a combination of a DC and an AC voltage.

To conduct a comparative analysis and avoid cumulative numerical errors, it is recommended to express Eqs. (5), 9, and 11 in dimensionless form [23]. This can be achieved by introducing the following dimensionless parameters and replacing them in the original equations:

$$\xi = \frac{x}{L_p}, \quad \hat{u}_2 = \frac{u_2}{L_p}, \quad \hat{\eta} = \frac{\eta}{h_p}, \quad \hat{w}_2 = \frac{w_2}{h_p}, \quad \hat{x} = \frac{x}{L}, \quad \hat{w} = \frac{w}{g_0}, \quad \hat{t} = \frac{t}{\sqrt{\frac{\rho A L^4}{EI}}} \quad (13)$$

The given equations result in the following coupled set of dimensionless nonlinear equations when these parameters are used:

$$\frac{\partial^2 \hat{u}_2}{\partial \hat{\xi}^2} + \alpha_1 \frac{\partial \hat{w}_2}{\partial \hat{\xi}} \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} + \alpha_5 \frac{\partial^3 \hat{u}_2}{\partial \hat{t} \partial \hat{\xi}^2} + \alpha_1 \alpha_5 \left(\frac{\partial^2 \hat{w}_2}{\partial \hat{t} \partial \hat{\xi}} \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} + \frac{\partial \hat{w}_2}{\partial \hat{\xi}} \frac{\partial^3 \hat{w}_2}{\partial \hat{t} \partial \hat{\xi}^2} \right) - \alpha_6 \frac{\partial^2 \hat{u}_2}{\partial \hat{t}^2} = \hat{f}_1^{(x)}(\hat{\xi}, \hat{t})$$

with

$$\begin{aligned} \hat{f}_1^{(x)}(\hat{\xi}, \hat{t}) = & -\alpha_2 \frac{\partial \hat{w}_2}{\partial \hat{\xi}} \hat{w}^{(x)} - \alpha_3 \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} \hat{w}^{(x)} - \alpha_4 \hat{w}^{2(x)} - \alpha_2 \alpha_5 \left(\frac{\partial^2 \hat{w}_2}{\partial \hat{t} \partial \hat{\xi}} \hat{w}^{(x)} + \frac{\partial \hat{w}_2}{\partial \hat{\xi}} \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} \right) \\ & - \alpha_3 \alpha_5 \left(\frac{\partial^3 \hat{w}_2}{\partial \hat{t} \partial \hat{\xi}^2} \hat{w}^{(x)} + \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} \right) - \alpha_4 \alpha_5 \hat{w}^{(x)} \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} + \alpha_7 \frac{\partial^2 \hat{w}^{(x)}}{\partial \hat{t}^2} \end{aligned} \quad (14)$$

$$\begin{aligned} & \frac{\partial^4 \hat{w}_2}{\partial \hat{\xi}^4} - \beta_1 \frac{\partial^2 \hat{u}_2}{\partial \hat{\xi}^2} \frac{\partial \hat{w}_2}{\partial \hat{\xi}} - \beta_1 \frac{\partial \hat{u}_2}{\partial \hat{\xi}} \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} - \beta_6 \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} \left(\frac{\partial \hat{w}_2}{\partial \hat{\xi}} \right)^2 + \beta_{12} \frac{\partial^5 \hat{w}_2}{\partial \hat{t} \partial \hat{\xi}^4} - \beta_1 \beta_{12} \left(\frac{\partial^3 \hat{u}_2}{\partial \hat{t} \partial \hat{\xi}^2} \frac{\partial \hat{w}_2}{\partial \hat{\xi}} + \frac{\partial^2 \hat{u}_2}{\partial \hat{\xi}^2} \frac{\partial^2 \hat{w}_2}{\partial \hat{t} \partial \hat{\xi}} \right) - \beta_1 \beta_{12} \left(\frac{\partial^2 \hat{u}_2}{\partial \hat{t} \partial \hat{\xi}} \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} + \frac{\partial \hat{u}_2}{\partial \hat{\xi}} \frac{\partial^3 \hat{w}_2}{\partial \hat{t} \partial \hat{\xi}^2} \right) \\ & - \beta_6 \beta_{12} \left(\frac{\partial^3 \hat{w}_2}{\partial \hat{t} \partial \hat{\xi}^2} \left(\frac{\partial \hat{w}_2}{\partial \hat{\xi}} \right)^2 + 2 \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} \frac{\partial \hat{w}_2}{\partial \hat{\xi}} \frac{\partial^2 \hat{w}_2}{\partial \hat{t} \partial \hat{\xi}} \right) + \beta_{13} \frac{\partial^2 \hat{w}_2}{\partial \hat{t}^2} \\ & = \hat{f}_2^{(x)}(\hat{\xi}, \hat{t}) \end{aligned} \quad (15)$$

with

$$\begin{aligned}
\hat{f}_2^{(x)}(\xi, \hat{t}) = & \beta_2 \frac{\partial^2 \hat{u}_2}{\partial \xi^2} \hat{w}^{(x)} + \beta_3 \frac{\partial \hat{u}_2}{\partial \xi} \hat{w}^{(x)} + \beta_4 \frac{\partial^2 \hat{w}_2}{\partial \xi^2} \hat{w}^{(x)} + \beta_5 (\hat{w}^{(x)})^2 + \beta_7 \frac{\partial \hat{w}_2}{\partial \xi} \frac{\partial^2 \hat{w}_2}{\partial \xi^2} \hat{w}^{(x)} + \beta_8 \frac{\partial^2 \hat{w}_2}{\partial \xi^2} (\hat{w}^{(x)})^2 + \beta_9 \left(\frac{\partial \hat{w}_2}{\partial \xi} \right)^2 \hat{w}^{(x)} + \beta_{10} \frac{\partial \hat{w}_2}{\partial \xi} (\hat{w}^{(x)})^2 \\
& + \beta_{11} (\hat{w}^{(x)})^3 + \beta_2 \beta_{12} \left(\frac{\partial^3 \hat{u}_2}{\partial \hat{t} \partial \xi^2} \hat{w}^{(x)} + \frac{\partial^2 \hat{u}_2}{\partial \xi^2} \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} \right) + \beta_3 \beta_{12} \left(\frac{\partial^2 \hat{u}_2}{\partial \hat{t} \partial \xi} \hat{w}^{(x)} + \frac{\partial \hat{u}_2}{\partial \xi} \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} \right) \\
& + \beta_4 \beta_{12} \left(\frac{\partial^3 \hat{w}_2}{\partial \hat{t} \partial \xi^2} \hat{w}^{(x)} + \frac{\partial^2 \hat{w}_2}{\partial \xi^2} \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} \right) + 2\beta_5 \beta_{12} \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} \hat{w}^{(x)} + \beta_7 \beta_{12} \left(\frac{\partial^2 \hat{w}_2}{\partial \hat{t} \partial \xi} \frac{\partial^2 \hat{w}_2}{\partial \xi^2} \hat{w}^{(x)} + \frac{\partial \hat{w}_2}{\partial \xi} \frac{\partial^3 \hat{w}_2}{\partial \hat{t} \partial \xi^2} \hat{w}^{(x)} + \frac{\partial \hat{w}_2}{\partial \xi} \frac{\partial^2 \hat{w}_2}{\partial \xi^2} \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} \right) \\
& - \beta_8 \beta_{12} \left(\frac{\partial^3 \hat{w}_2}{\partial \hat{t} \partial \xi^2} (\hat{w}^{(x)})^2 + 2 \frac{\partial^2 \hat{w}_2}{\partial \xi^2} \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} \hat{w}^{(x)} \right) \\
& + \beta_9 \beta_{12} \left(2 \frac{\partial \hat{w}_2}{\partial \xi} \frac{\partial^2 \hat{w}_2}{\partial \hat{t} \partial \xi} \hat{w}^{(x)} + \left(\frac{\partial \hat{w}_2}{\partial \xi} \right)^2 \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} \right) + \beta_{10} \beta_{12} \left(\frac{\partial^2 \hat{w}_2}{\partial \hat{t} \partial \xi} (\hat{w}^{(x)})^2 + 2 \frac{\partial \hat{w}_2}{\partial \xi} \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} \hat{w}^{(x)} \right) + 3\beta_{11} \beta_{12} \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} (\hat{w}^{(x)})^2 - \beta_{14} \frac{\partial^2 \hat{w}^{(x)}}{\partial \hat{t}^2}
\end{aligned}$$

and

$$\frac{\partial^4 \hat{w}}{\partial \hat{x}^4} - \gamma_1 \left(\int_0^1 \left(\frac{\partial \hat{w}}{\partial \hat{x}} \right)^2 d\hat{x} \right) \frac{\partial^2 \hat{w}}{\partial \hat{x}^2} + \frac{\partial^2 \hat{w}}{\partial \hat{t}^2} + \gamma_2 \gamma_4 \frac{\partial \hat{w}}{\partial \hat{t}} - \gamma_2 \gamma_6 \frac{\partial \hat{w}}{\partial \hat{t}} = \hat{f}_3^{(x)}(\hat{t}) + \gamma_7 \hat{f}_e(\hat{x}, \hat{t}) \quad (16)$$

with

$$\hat{f}_3^{(x)}(\hat{t}) = \gamma_3 \frac{\partial \hat{u}_2}{\partial \xi} \Big|_{\xi=1} - \gamma_4 \hat{w} - \gamma_5 \frac{\partial^3 \hat{w}_2}{\partial \xi^3} \Big|_{\xi=1} + \gamma_6 \hat{w} - \gamma_2 \gamma_3 \frac{\partial^2 \hat{u}_2}{\partial \hat{t} \partial \xi} \Big|_{\xi=1} - \gamma_2 \gamma_5 \frac{\partial^4 \hat{w}_2}{\partial \hat{t} \partial \xi^3} \Big|_{\xi=1}$$

with the following coefficients:

$$\begin{aligned}
\alpha_1 = & \left(\frac{h_p}{L_p} \right)^2, \alpha_2 = 6g_0 \frac{h_p}{L_p^2} (1 - 2\hat{\xi}) \cos\theta, \alpha_3 = 6g_0 \frac{h_p}{L_p^2} (\hat{\xi} - \hat{\xi}^2) \cos\theta, \alpha_4 = 36 \frac{g_0^2}{L_p^3} (1 - 2\hat{\xi}) (\hat{\xi} - \hat{\xi}^2) \cos^2\theta, \alpha_5 = \frac{\tau}{t^*}, \alpha_6 = \frac{\rho_p L_p^2}{E_p t^{*2}}, \alpha_7 \\
= & \frac{g_0 L_p \rho_p}{E_p t^{*2}} \hat{\xi} \sin\theta, \\
\beta_1 = & \frac{A_p L_p^2}{I_p}, \beta_2 = 6(\hat{\xi} - \hat{\xi}^2) \frac{A_p L_p^2 g_0}{I_p h_p} \cos\theta, \beta_3 = 6(1 - 2\hat{\xi}) \frac{A_p L_p^2 g_0}{I_p h_p} \cos\theta, \beta_4 = -\frac{A_p L_p g_0}{I_p} \sin\theta, \\
\beta_5 = & -6(1 - 2\hat{\xi}) \frac{A_p L_p g_0^2}{I_p h_p} \sin\theta \cos\theta, \beta_6 = \frac{3}{2} \frac{A_p h_p^2}{I_p}, \beta_7 = 18 \frac{A_p h_p g_0}{I_p} (\hat{\xi} - \hat{\xi}^2) \cos\theta, \beta_8 = 54 \frac{A_p g_0^2}{I_p} (\hat{\xi} - \hat{\xi}^2)^2 \cos^2\theta, \beta_9 \\
= & 9 \frac{A_p h_p g_0}{I_p} (1 - 2\hat{\xi}) \cos\theta, \\
\beta_{10} = & 108 \frac{A_p g_0^2}{I_p} (1 - 2\hat{\xi}) (\hat{\xi} - \hat{\xi}^2) \cos^2\theta, \beta_{11} = 324 \frac{A_p g_0^3}{I_p h_p} (1 - 2\hat{\xi}) (\hat{\xi} - \hat{\xi}^2)^2 \cos^3\theta, \\
\beta_{12} = & \frac{\tau}{t^*}, \beta_{13} = \frac{\rho_p A_p L_p^4}{E_p I_p t^*}, \beta_{14} = \frac{\rho_p A_p L_p^4 g_0}{E_p I_p h_p t^{*2}} (3\hat{\xi}^2 - 2\hat{\xi}^3) \cos\theta, \\
\gamma_1 = & \frac{A g_0^2 (1 - \nu^2)}{2I}, \gamma_2 = \frac{\tau}{t^*}, \gamma_3 = N_p \frac{E_p A_p L^3 (1 - \nu^2)}{E I g_0} \sin\theta, \gamma_4 = N_p \frac{E_p A_p L^3 (1 - \nu^2)}{E I L_p} \sin^2\theta, \\
\gamma_5 = & N_p \frac{E_p I_p h_p L^3 (1 - \nu^2)}{E I g_0 L_p^3} \cos\theta, \gamma_6 = N_p \frac{12 E_p I_p L^3 (1 - \nu^2)}{E I L_p^3} \cos^2\theta, \gamma_7 = \frac{L^4 (1 - \nu^2)}{E I g_0}
\end{aligned} \quad (17)$$

It should be noted that the equations that govern the static deflection of the microbeam under DC voltage can be written by removing the time-dependent parameters from Eqs. (14)-16, as following:

$$\frac{\partial^2 \hat{u}_2}{\partial \hat{\xi}^2} + \alpha_1 \frac{\partial \hat{w}_2}{\partial \hat{\xi}} \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} = \hat{f}_1^{(x)}(\hat{\xi})$$

with

$$\hat{f}_1^{(i)}(\hat{\xi}) = -\alpha_2 \frac{\partial \hat{w}_2}{\partial \hat{\xi}} \hat{w}^{(x)} - \alpha_3 \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} \hat{w}^{(x)} - \alpha_4 (\hat{w}^{(x)})^2, \quad (18)$$

$$\frac{\partial^4 \hat{w}_2}{\partial \hat{\xi}^4} - \beta_1 \frac{\partial^2 u_2}{\partial \hat{\xi}^2} \frac{\partial \hat{w}_2}{\partial \hat{\xi}} - \beta_1 \frac{\partial \hat{u}_2}{\partial \hat{\xi}} \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} - \beta_6 \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} \left(\frac{\partial \hat{w}_2}{\partial \hat{\xi}} \right)^2 = \hat{f}_2^{(i)}(\hat{\xi}) \quad (19)$$

with

$$\begin{aligned} \hat{f}_2^{(i)}(\hat{\xi}) = & \beta_2 \frac{\partial^2 u_2}{\partial \hat{\xi}^2} \hat{w}^{(x)} + \beta_3 \frac{\partial \hat{u}_2}{\partial \hat{\xi}} \hat{w}^{(x)} + \beta_4 \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} \hat{w}^{(x)} + \beta_5 (\hat{w}^{(x)})^2 + \beta_7 \frac{\partial \hat{w}_2}{\partial \hat{\xi}} \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} \hat{w}^{(x)} + \beta_8 \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} (\hat{w}^{(x)})^2 + \beta_9 \left(\frac{\partial \hat{w}_2}{\partial \hat{\xi}} \right)^2 \hat{w}^{(x)} + \beta_{10} \frac{\partial \hat{w}_2}{\partial \hat{\xi}} (\hat{w}^{(x)})^2 \\ & + \beta_{11} (\hat{w}^{(x)})^3 \end{aligned}$$

and

$$\frac{\partial^4 \hat{w}}{\partial \hat{x}^4} - \gamma_1 \left(\int_0^1 \left(\frac{\partial \hat{w}}{\partial \hat{x}} \right)^2 d\hat{x} \right) \frac{\partial^2 \hat{w}}{\partial \hat{x}^2} = \hat{f}_3^{(i)} + \gamma_7 \hat{f}_e(\hat{x}, \hat{t}) \quad (20)$$

with

$$\hat{f}_3^{(i)} = -\gamma_3 \frac{\partial \hat{u}_2}{\partial \hat{\xi}} \bigg|_{\hat{\xi}=1} - \gamma_4 \hat{w} - \gamma_5 \frac{\partial^3 \hat{w}_2}{\partial \hat{\xi}^3} \bigg|_{\hat{\xi}=1} + \gamma_6 \hat{w}$$

3. Solution method

3.1. Static analysis

Electrostatic MEMS devices typically use a DC voltage for bias, tuning, or switching purposes. The governing static equations must be solved numerically due to their nonlinearity. In this case, the step-by-step linearization method (SSLM) is used as a powerful technique [24]. This method considers the gradual increase of the applied voltage (dV) and calculates the resulting deflections at each step ($\hat{X}(\hat{\xi})$, $\hat{\Psi}(\hat{\xi})$ and $\hat{\Phi}(\hat{x})$, respectively for longitudinal and bending deformation of pillars and bending deformation of the micro-beam). The resulting deflections can be estimated by using the Hilbert space basis functions, and then the Galerkin weighted residual method (respectively with the weight functions $\chi(\hat{\xi})$, $\psi(\hat{\xi})$ and $\varphi(\hat{x})$) are used to obtain the following linear set of algebraic equations:

$$\begin{cases} K_{11}^s q_1 + K_{12}^s q_2 = F_1^s \\ K_{21}^s q_1 + K_{22}^s q_2 = F_2^s \\ K_{33}^s q_3 = F_3^s \end{cases} \quad (21)$$

where

$$K_{11}^s = \int_0^1 \hat{\chi}(\hat{\xi}) \frac{\partial^2 \hat{\chi}}{\partial \hat{\xi}^2} d\hat{\xi} \quad (22)$$

$$K_{12}^s = \int_0^1 \alpha_1 \hat{\chi}(\hat{\xi}) \left(\frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} \bigg|_{(k)} \frac{\partial \hat{\psi}}{\partial \hat{\xi}} + \frac{\partial \hat{w}_2}{\partial \hat{\xi}} \bigg|_{(k)} \frac{\partial^2 \hat{\psi}}{\partial \hat{\xi}^2} \right) d\hat{\xi},$$

$$K_{21}^s = \int_0^1 -\beta_1 \hat{\psi}(\hat{\xi}) \left(\frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} \bigg|_{(k)} \frac{\partial \hat{\chi}}{\partial \hat{\xi}} + \frac{\partial \hat{w}_2}{\partial \hat{\xi}} \bigg|_{(k)} \frac{\partial^2 \hat{\chi}}{\partial \hat{\xi}^2} \right) d\hat{\xi},$$

$$K_{22}^s = \int_0^1 \hat{\psi}(\hat{\xi}) \left(\frac{\partial^4 \hat{\psi}}{\partial \hat{\xi}^4} - \left(\beta_1 \frac{\partial^2 \hat{u}_2}{\partial \hat{\xi}^2} \bigg|_{(k)} + 2\beta_6 \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} \bigg|_{(k)} \frac{\partial \hat{\psi}}{\partial \hat{\xi}} - \left(\beta_1 \frac{\partial \hat{u}_2}{\partial \hat{\xi}} \bigg|_{(k)} + \beta_6 \left(\frac{\partial \hat{w}_2}{\partial \hat{\xi}} \right)^2 \bigg|_{(k)} \right) \frac{\partial^2 \hat{\psi}}{\partial \hat{\xi}^2} \right) d\hat{\xi},$$

$$K_{33}^s = \int_0^1 \hat{\varphi}(\hat{x}) \left(\frac{\partial^4 \hat{\varphi}}{\partial \hat{x}^4} - \gamma_1 \left(2 \frac{\partial^2 \hat{w}}{\partial \hat{x}^2} \bigg|_{(k)} \left(\int_0^1 \frac{\partial \hat{w}}{\partial \hat{x}} \bigg|_{(k)} \frac{\partial \hat{\varphi}}{\partial \hat{x}} d\hat{x} \right) + \left(\int_0^1 \left(\frac{\partial \hat{w}}{\partial \hat{x}} \right)^2 \bigg|_{(k)} d\hat{x} \right) \frac{\partial^2 \hat{\varphi}}{\partial \hat{x}^2} - \left(2\gamma_7 \frac{V^2}{(1 - \hat{w})^3} \bigg|_{(k)} \right) \hat{\varphi} \right) d\hat{x},$$

$$F_1^s = \int_0^1 \widehat{\chi}(\xi) \left(-\alpha_2 \frac{\partial \widehat{w}_2}{\partial \xi} \widehat{w}^{(x)} - \alpha_3 \frac{\partial^2 \widehat{w}_2}{\partial \xi^2} \widehat{w}^{(x)} - \alpha_4 \widehat{w}^{2(x)} \right) \Big|_k d\xi,$$

$$F_2^s = \int_0^1 \widehat{\psi}(\xi) \left(\beta_2 \frac{\partial^2 u_2}{\partial \xi^2} \widehat{w}^{(x)} + \beta_3 \frac{\partial \widehat{w}_2}{\partial \xi} \widehat{w}^{(x)} + \beta_4 \frac{\partial^2 \widehat{w}_2}{\partial \xi^2} \widehat{w}^{(x)} + \beta_5 (\widehat{w}^{(x)})^2 + \beta_7 \frac{\partial \widehat{w}_2}{\partial \xi} \frac{\partial^2 \widehat{w}_2}{\partial \xi^2} \widehat{w}^{(x)} + \beta_8 \frac{\partial^2 \widehat{w}_2}{\partial \xi^2} (\widehat{w}^{(x)})^2 \right. \\ \left. + \beta_9 \left(\frac{\partial \widehat{w}_2}{\partial \xi} \right)^2 \widehat{w}^{(x)} + \beta_{10} \frac{\partial \widehat{w}_2}{\partial \xi} (\widehat{w}^{(x)})^2 + \beta_{11} (\widehat{w}^{(x)})^3 \right) \Big|_k d\xi$$

and

$$F_3^s = \int_0^1 \widehat{\varphi}(\widehat{x}) \left(-\gamma_3 \frac{\partial \widehat{X}}{\partial \xi} \Big|_{\xi=1} - \gamma_4 \widehat{\Phi} - \gamma_5 \frac{\partial^3 \widehat{\Psi}}{\partial \xi^3} \Big|_{\xi=1} + \gamma_6 \widehat{\Phi} + 2\gamma_7 \frac{V}{(1 - \widehat{w})^2} \Big|_{(k)} \right) d\widehat{x}$$

The terms indexed by k represent the values from the previous step in each current step of SSLM. $\chi_i(\xi)$, $\psi_i(\xi)$ and $\varphi_i(\widehat{x})$ are shape functions that satisfy the relevant boundary conditions. It is important to note that selecting small enough voltage steps will result in accurate predictions of deformations using only the first term of each shape function. The unknown coefficients of Eq. (21) can be calculated by solving this set, which in turn allows for the determination of the nonlinear static deflections of the system's elements.

3.2. Dynamic analysis

To address the system behavior under dynamic actuations such as step DC or AC voltages, the Eqs. (14)–16 should be solved. The required numerical solution can be started by transforming them into a reduced order model using the Hilbert space basis functions as follows:

$$\widehat{u}_2^{(Q)}(\xi, \widehat{t}) = \sum_{i=1}^Q \chi_i(\xi) q_{1i}(\widehat{t}) \quad (23)$$

$$\widehat{w}_2^{(R)}(\xi, \widehat{t}) = \sum_{i=1}^R \psi_i(\xi) q_{2i}(\widehat{t})$$

$$\widehat{w}^{(S)}(\widehat{x}, \widehat{t}) = \sum_{i=1}^S \varphi_i(\widehat{x}) q_{3i}(\widehat{t})$$

Substituting each of these functions into Eqs. (14)–16 gives the following set of equations:

$$\sum_{i=1}^Q \chi_i'(\xi) q_{1i}(\widehat{t}) + \alpha_1 \sum_{i=1}^R \sum_{j=1}^R \psi_i'(\xi) \psi_j'(\xi) q_{2i}(\widehat{t}) q_{2j}(\widehat{t}) + \alpha_5 \sum_{i=1}^Q \chi_i'(\xi) \dot{q}_{1i}(\widehat{t}) \\ + \alpha_1 \alpha_5 \sum_{i=1}^R \sum_{j=1}^R \left(\psi_i'(\xi) \psi_j'(\xi) \dot{q}_{2i}(\widehat{t}) q_{2j}(\widehat{t}) + \psi_i'(\xi) \psi_j'(\xi) q_{2i}(\widehat{t}) \dot{q}_{2j}(\widehat{t}) \right) - \alpha_6 \sum_{i=1}^Q \chi_i(\xi) \ddot{q}_{1i}(\widehat{t}) = \widehat{f}_1^{(k)}(\xi, \widehat{t}), \quad (24)$$

$$\sum_{i=1}^R \psi_i^{iv}(\xi) q_{2i}(\widehat{t}) - \beta_1 \sum_{i=1}^Q \sum_{j=1}^Q \chi_i'(\xi) \psi_j'(\xi) q_{1i}(\widehat{t}) q_{2j}(\widehat{t}) - \beta_1 \sum_{i=1}^Q \sum_{j=1}^R \chi_i'(\xi) \psi_j'(\xi) q_{1i}(\widehat{t}) q_{2j}(\widehat{t}) \\ - \beta_6 \sum_{i=1}^R \sum_{j=1}^R \sum_{k=1}^R \psi_i'(\xi) \psi_j'(\xi) \psi_k'(\xi) q_{2i}(\widehat{t}) q_{2j}(\widehat{t}) q_{2k}(\widehat{t}) + \beta_{12} \sum_{i=1}^R \psi_i^{iv}(\xi) \dot{q}_{2i}(\widehat{t}) \\ - \beta_1 \beta_{12} \sum_{i=1}^Q \sum_{j=1}^R \left(\chi_i'(\xi) \psi_j'(\xi) \dot{q}_{1i}(\widehat{t}) q_{2j}(\widehat{t}) + \chi_i'(\xi) \psi_j'(\xi) q_{1i}(\widehat{t}) \dot{q}_{2j}(\widehat{t}) \right) \\ - \beta_1 \beta_{12} \sum_{i=1}^Q \sum_{j=1}^R \left(\chi_i'(\xi) \psi_j'(\xi) \dot{q}_{1i}(\widehat{t}) q_{2i}(\widehat{t}) + \chi_i'(\xi) \psi_j'(\xi) q_{1i}(\widehat{t}) \dot{q}_{2j}(\widehat{t}) \right) \\ - \beta_6 \beta_{12} \sum_{i=1}^R \sum_{j=1}^R \sum_{k=1}^R \left(\psi_i'(\xi) \psi_j'(\xi) \psi_k'(\xi) \dot{q}_{2i}(\widehat{t}) q_{2j}(\widehat{t}) q_{2k}(\widehat{t}) + 2\psi_i'(\xi) \psi_j'(\xi) \psi_k'(\xi) q_{2i}(\widehat{t}) q_{2j}(\widehat{t}) \dot{q}_{2k}(\widehat{t}) \right) \\ + \beta_{13} \sum_{i=1}^R \psi_i(\xi) \ddot{q}_{2i}(\widehat{t}) = \widehat{f}_2^{(k)}(\xi, \widehat{t}) \quad (25)$$

and

$$\begin{aligned} & \sum_{i=1}^S \varphi_i^{iv}(\hat{x}) q_{3_i}(\hat{t}) - \gamma_1 \sum_{i=1}^S \sum_{j=1}^S \sum_{k=1}^S \left(\int_0^1 \varphi_i'(\hat{x}) \varphi_j'(\hat{x}) d\hat{x} \right) \varphi_k'(\hat{x}) q_{3_i}(\hat{t}) q_{3_j}(\hat{t}) q_{3_k}(\hat{t}) \\ & + \sum_{i=1}^S \varphi_i(\hat{x}) \ddot{q}_{3_i}(\hat{t}) + \gamma_2(\gamma_4 - \gamma_6) \sum_{i=1}^S \varphi_i(\hat{x}) \dot{q}_{3_i}(\hat{t}) = \widehat{f}_3^{(x)}(\hat{t}) + \gamma_7 \widehat{f}_e(\hat{x}, \hat{t}) \end{aligned} \quad (26)$$

Now, multiplying Eqs. (24)–26 by $\chi_i(\hat{\xi})$, $\psi_i(\hat{\xi})$ and $\varphi_i(\hat{x})$, respectively, and integrating the results from 0 to 1, the following three sets of ordinary differential equations are obtained:

$$\begin{aligned} & \sum_{j=1}^Q M_{11ij} \ddot{q}_{1_j}(\hat{t}) + \sum_{j=1}^Q C_{11ij} \dot{q}_{1_j}(\hat{t}) + \sum_{j=1}^Q K_{11ij} q_{1_j}(\hat{t}) \\ & + \sum_{k=1}^R \sum_{j=1}^R K'_{12jk} q_{2_j}(\hat{t}) q_{2_k}(\hat{t}) \\ & + \sum_{k=1}^R \sum_{j=1}^R H_{22jk} \dot{q}_{2_j}(\hat{t}) q_{2_k}(\hat{t}) = \left(\widehat{\mathcal{F}}_1^{(x)} \right)_i, \quad i = 1, 2, \dots, Q \\ & \sum_{j=1}^R M_{22ij} \ddot{q}_{2_j}(\hat{t}) + \sum_{j=1}^R C_{22ij} \dot{q}_{2_j}(\hat{t}) + \sum_{j=1}^R K_{22ij} q_{2_j}(\hat{t}) + \sum_{l=1}^R \sum_{k=1}^R \sum_{j=1}^R K''_{22jkl} q_{2_j}(\hat{t}) q_{2_k}(\hat{t}) q_{2_l}(\hat{t}) \\ & + \sum_{k=1}^R \sum_{j=1}^Q I_{12jk} q_{1_j}(\hat{t}) q_{2_k}(\hat{t}) + \sum_{k=1}^R \sum_{j=1}^Q I_{12jk} \dot{q}_{1_j}(\hat{t}) q_{2_k}(\hat{t}) + \sum_{k=1}^R \sum_{j=1}^Q I_{12jk} q_{1_j}(\hat{t}) \dot{q}_{2_k}(\hat{t}) \\ & + \sum_{l=1}^R \sum_{k=1}^R \sum_{j=1}^R I_{22jkl} \dot{q}_{2_j}(\hat{t}) q_{2_k}(\hat{t}) q_{2_l}(\hat{t}) = \left(\widehat{\mathcal{F}}_2^{(x)} \right)_i, \quad i = 1, 2, \dots, R \\ & \sum_{j=1}^S M_{33ij} \ddot{q}_{3_j}(\hat{t}) + \sum_{j=1}^S C_{33ij} \dot{q}_{3_j}(\hat{t}) + \sum_{j=1}^S K_{33ij} q_{3_j}(\hat{t}) + \sum_{l=1}^S \sum_{k=1}^S \sum_{j=1}^S K''_{33jkl} q_{3_j}(\hat{t}) q_{3_k}(\hat{t}) q_{3_l}(\hat{t}) = \widehat{\mathcal{F}}_{3_i} + \gamma_7 \widehat{F}_{e_i}(\hat{x}, \hat{t}), \quad i = 1, 2, \dots, S \end{aligned} \quad (27)$$

where

$$\begin{aligned} M_{11ij} &= - \int_0^1 \alpha_6 \chi_i(\hat{\xi}) \chi_j(\hat{\xi}) d\hat{\xi}, \\ C_{11ij} &= \int_0^1 \alpha_5 \chi_i(\hat{\xi}) \chi_j''(\hat{\xi}) d\hat{\xi}, \\ K_{11ij} &= \int_0^1 \chi_i(\hat{\xi}) \chi_j''(\hat{\xi}) d\hat{\xi}, \\ K'_{12ijk} &= \int_0^1 \alpha_1 \chi_i(\hat{\xi}) \psi_j'(\hat{\xi}) \psi_k''(\hat{\xi}) d\hat{\xi}, \\ H_{22ijk} &= \int_0^1 2\alpha_1 \alpha_5 \chi_i(\hat{\xi}) \psi_j'(\hat{\xi}) \psi_k''(\hat{\xi}) d\hat{\xi}, \\ \left(\widehat{\mathcal{F}}_1^{(x)} \right)_i &= \int_0^1 \chi_i(\hat{\xi}) \left(-\alpha_2 \frac{\partial \widehat{w}_2^{(x)}}{\partial \hat{\xi}} - \alpha_3 \frac{\partial^2 \widehat{w}_2^{(x)}}{\partial \hat{\xi}^2} - \alpha_4 \widehat{w}_2^{(x)} - \alpha_2 \alpha_5 \left(\frac{\partial^2 \widehat{w}_2^{(x)}}{\partial \hat{t} \partial \hat{\xi}} + \frac{\partial \widehat{w}_2^{(x)}}{\partial \hat{\xi}} \frac{\partial \widehat{w}^{(x)}}{\partial \hat{t}} \right) - \alpha_3 \alpha_5 \left(\frac{\partial^3 \widehat{w}_2^{(x)}}{\partial \hat{t} \partial \hat{\xi}^2} \widehat{w}^{(x)} + \frac{\partial^2 \widehat{w}_2^{(x)}}{\partial \hat{\xi}^2} \frac{\partial \widehat{w}^{(x)}}{\partial \hat{t}} \right) - \alpha_4 \alpha_5 \widehat{w}^{(x)} \frac{\partial \widehat{w}^{(x)}}{\partial \hat{t}} + \alpha_7 \frac{\partial^2 \widehat{w}^{(x)}}{\partial \hat{t}^2} \right) d\hat{\xi}, \end{aligned} \quad (28)$$

$$M_{22ij} = \int_0^1 \beta_{13} \psi_i(\hat{\xi}) \psi_j(\hat{\xi}) d\hat{\xi}, \quad (29)$$

$$C_{22ij} = \int_0^1 \beta_{12} \psi_i(\hat{\xi}) \psi_j^{iv}(\hat{\xi}) d\hat{\xi},$$

$$K_{22ij} = \int_0^1 \psi_i(\hat{\xi}) \psi_j^{iv}(\hat{\xi}) d\hat{\xi},$$

$$K_{22}^{*ijkl} = \int_0^1 -\beta_6 \psi_i(\hat{\xi}) \psi_j'(\hat{\xi}) \psi_k'(\hat{\xi}) \psi_l''(\hat{\xi}) d\hat{\xi},$$

$$I_{12ijk} = \int_0^1 -\beta_1 \psi_i(\hat{\xi}) (\chi_j''(\hat{\xi}) \psi_k'(\hat{\xi}) + \chi_j'(\hat{\xi}) \psi_k''(\hat{\xi})) d\hat{\xi},$$

$$I_{12}^{*ijk} = \int_0^1 -\beta_1 \beta_{12} \psi_i(\hat{\xi}) (\chi_j''(\hat{\xi}) \psi_k'(\hat{\xi}) + \chi_j'(\hat{\xi}) \psi_k''(\hat{\xi})) d\hat{\xi},$$

$$I_{12}^{*ijk} = \int_0^1 -\beta_1 \beta_{12} \psi_i(\hat{\xi}) (\chi_j''(\hat{\xi}) \psi_k'(\hat{\xi}) + \chi_j'(\hat{\xi}) \psi_k''(\hat{\xi})) d\hat{\xi},$$

$$I_{222}^{*ijkl} = \int_0^1 -3\beta_6 \beta_{12} \psi_i(\hat{\xi}) \psi_j'(\hat{\xi}) \psi_k'(\hat{\xi}) \psi_l''(\hat{\xi}) d\hat{\xi},$$

$$\begin{aligned} (F_2^{(x)})_i &= \int_0^1 \psi_i(\hat{\xi}) \left(\beta_2 \frac{\partial^2 u_2}{\partial \hat{\xi}^2} \hat{w}^{(x)} + \beta_3 \frac{\partial \hat{u}_2}{\partial \hat{\xi}} \hat{w}^{(x)} + \beta_4 \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} \hat{w}^{(x)} + \beta_5 (\hat{w}^{(x)})^2 + \beta_7 \frac{\partial \hat{w}_2}{\partial \hat{\xi}} \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} \hat{w}^{(x)} \right. \\ &+ \beta_8 \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} (\hat{w}^{(x)})^2 + \beta_9 \left(\frac{\partial \hat{w}_2}{\partial \hat{\xi}} \right)^2 \hat{w}^{(x)} + \beta_{10} \frac{\partial \hat{w}_2}{\partial \hat{\xi}} (\hat{w}^{(x)})^2 + \beta_{11} (\hat{w}^{(x)})^3 \\ &+ \beta_2 \beta_{12} \left(\frac{\partial^3 u_2}{\partial \hat{t} \partial \hat{\xi}^2} \hat{w}^{(x)} + \frac{\partial^2 u_2}{\partial \hat{\xi}^2} \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} \right) + \beta_3 \beta_{12} \left(\frac{\partial^2 \hat{u}_2}{\partial \hat{t} \partial \hat{\xi}} \hat{w}^{(x)} + \frac{\partial \hat{u}_2}{\partial \hat{\xi}} \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} \right) \\ &+ \beta_4 \beta_{12} \left(\frac{\partial^3 \hat{w}_2}{\partial \hat{t} \partial \hat{\xi}^2} \hat{w}^{(x)} + \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} \right) + 2\beta_5 \beta_{12} \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} \hat{w}^{(x)} \\ &+ \beta_7 \beta_{12} \left(\frac{\partial^2 \hat{w}_2}{\partial \hat{t} \partial \hat{\xi}} \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} \hat{w}^{(x)} + \frac{\partial \hat{w}_2}{\partial \hat{\xi}} \frac{\partial^3 \hat{w}_2}{\partial \hat{t} \partial \hat{\xi}^2} \hat{w}^{(x)} + \frac{\partial \hat{w}_2}{\partial \hat{\xi}} \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} \right) \\ &- \beta_8 \beta_{12} \left(\frac{\partial^3 \hat{w}_2}{\partial \hat{t} \partial \hat{\xi}^2} (\hat{w}^{(x)})^2 + 2 \frac{\partial^2 \hat{w}_2}{\partial \hat{\xi}^2} \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} \hat{w}^{(x)} \right) + \beta_9 \beta_{12} \left(2 \frac{\partial \hat{w}_2}{\partial \hat{\xi}} \frac{\partial^2 \hat{w}_2}{\partial \hat{t} \partial \hat{\xi}} \hat{w}^{(x)} + \left(\frac{\partial \hat{w}_2}{\partial \hat{\xi}} \right)^2 \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} \right) \\ &+ \beta_{10} \beta_{12} \left(\frac{\partial^2 \hat{w}_2}{\partial \hat{t} \partial \hat{\xi}} (\hat{w}^{(x)})^2 + 2 \frac{\partial \hat{w}_2}{\partial \hat{\xi}} \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} \hat{w}^{(x)} \right) + 3\beta_{11} \beta_{12} \frac{\partial \hat{w}^{(x)}}{\partial \hat{t}} (\hat{w}^{(x)})^2 - \beta_{14} \frac{\partial^2 \hat{w}^{(x)}}{\partial \hat{t}^2} \hat{w}^{(x)} \Big) d\hat{\xi} \end{aligned}$$

and

$$M_{33ij} = \int_0^1 \varphi_i(\hat{x}) \varphi_j(\hat{x}) d\hat{x}, \quad C_{33ij} = \int_0^1 \gamma_2 (\gamma_4 - \gamma_6) \varphi_i(\hat{x}) \varphi_j(\hat{x}) d\hat{x}, \quad (30)$$

$$K_{33ij} = \int_0^1 \varphi_i(\hat{x}) \varphi_j^{iv}(\hat{x}) d\hat{x}, \quad K_{33}^{*ijkl} = -\gamma_1 \int_0^1 \varphi_i(\hat{x}) \varphi_j'(\hat{x}) \left(\int_0^1 \varphi_k(\hat{x}) \varphi_l(\hat{x}) d\hat{x} \right) d\hat{x},$$

$$\widehat{\mathcal{F}}_{3i} = \int_0^1 \varphi_i(\widehat{x}) \left(-\gamma_3 \frac{\partial \widehat{u}_2}{\partial \widehat{\xi}} \bigg|_{\widehat{\xi}=1} - \gamma_4 \widehat{w} - \gamma_5 \frac{\partial^3 \widehat{w}_2}{\partial \widehat{\xi}^3} \bigg|_{\widehat{\xi}=1} + \gamma_6 \widehat{w} - \gamma_2 \gamma_3 \frac{\partial^2 \widehat{u}_2}{\partial t \partial \widehat{\xi}} \bigg|_{\widehat{\xi}=1} - \gamma_2 \gamma_5 \frac{\partial^4 \widehat{w}_2}{\partial t \partial \widehat{\xi}^3} \bigg|_{\widehat{\xi}=1} + \gamma_7 \widehat{F}_e(\widehat{x}, \widehat{t}) \right) d\widehat{x}$$

Now, the resulted linear differential equations set of (27) can be integrated over time using various numerical methods, such as the Runge-Kutta method, to determine the vibrational response of the system's components.

4. Results and discussion

To validate the numerical procedures for the proposed static model, the results obtained for a traditional capacitive system (air gap) are compared with those reported in the literature. Osterberg and Senturia [25] have reported the pull-in voltage for a given fixed-fixed capacitive microbeam as 20.2 and 20.3 V by energy and MEMCAD methods, respectively. The calculated pull-in voltage by SSM is very close to their results at 20.1 V. The pull-in instability is a discontinuity that arises from the interplay between elastic and electrostatic forces. When a potential difference is applied between a conducting structure and ground level, the structure deforms due to electrostatic forces. The elastic forces increase linearly with displacement, while the electrostatic forces decrease inversely proportional to the square of the distance. When the voltage is increased, the displacement increases until the electrostatic force growth rate exceeds that of the elastic force, resulting in a force imbalance and physical contact, leading to pull-in instability. This phenomenon is known as the static pull-in voltage [24].

To validate the dynamic results with previous works, we consider an air-gap electrostatic fixed-fixed microbeam with the same specifications as the pressure sensor used by Hung and Senturia [26]. Fig. 3 compares the pull-in time calculated using our proposed method with the theoretical and experimental results of Hung and Senturia for various values of step DC voltage. The pull-in time is determined by monitoring the beam response over time for a sudden increase in displacement. At this point, the time is recorded as the pull-in time [26]. As shown in Fig. 2, the calculated results are in excellent agreement with the theoretical and experimental results. It is demonstrated that the pull-in instability does not occur before $V = 8.18$ V in the no damping case. Therefore, this step DC voltage can be introduced as the dynamic pull-in voltage for the microbeam.

The following section addresses the static and dynamic behavior of the proposed system using the provided mathematical model. Table 1 displays the material properties and geometries of the model under consideration (see Fig. 1). It is important to note that the microbeam and pillars are assumed to be made of silicon and PDMS materials, respectively.

Fig. 4 shows the variation in the deflection of the microbeam center point with DC voltage. The inclusion of the PDMS gap leads to a significant increase in beam deformation for a given voltage. In addition, tilting the pillars further increases the deformation. The former is due to the increase in the equivalent dielectric constant of the gap medium with PDMS, which directly increases the electrostatic force generated for a given voltage, and the latter is due to the dominance of the bending deformation of the tilted pillars rather than the compressive deformation of the vertical pillars. It is noted that the ratio of the total surface area of pillars to the surface area of the beam are considered to be 0.3 here.

Fig. 5 illustrates the transverse and longitudinal deformations of the central pillar. Parts (a) and (b) are plotted for a 25 V actuation voltage and different tilt angles, while parts (c) and (d) are plotted for a $\theta = \frac{\pi}{4}$ with different actuation voltages. The figure illustrates that increasing the tilt angle of the pillars results in an increase in longitudinal deformations and a decrease in transverse deformation. This is because the longitudinal deformation dominates for tilt angles bigger than $\frac{\pi}{4}$. Furthermore, increasing the voltage significantly increases the deformations for a given angle. An interesting observation of this figure is the dominant nonlinear functionality of the transverse deformation compared to the longitudinal deformation for the given range of actuation. Such a result is obtained for the different pillars that are connected in different positions to the beam. As shown in Fig. 6, the deformations of the pillars increase significantly when the connection points are approaching from the beam anchors to the centre.

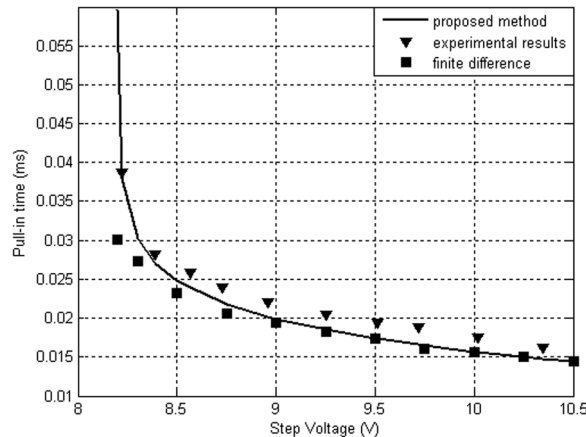


Fig. 3. Comparison of the obtained pull-in time for an air gap system with the reported data.

Table 1
Material properties and geometries of the system.

Physical parameters	Symbols		Values		units
	beam	pillar	beam	pillar	
Young's modulus	E	E_p	169	0.0005	GPa
Poisson's ratio	ν	ν_p	0.06	0.5	–
Density	ρ	ρ_p	2331	970	Kg/m ³
Length	L	L_p	200	dependent on g_0 and θ	μm
Width	b	b_p	20	0.15	μm
Thickness	h	h_p	1.5	0.15	μm
Initial gap	g_0	–	1.5	–	μm
Dielectric constant	–	ϵ_p	–	2.98	–
Pillar's angle	–	θ	–	variable	rad

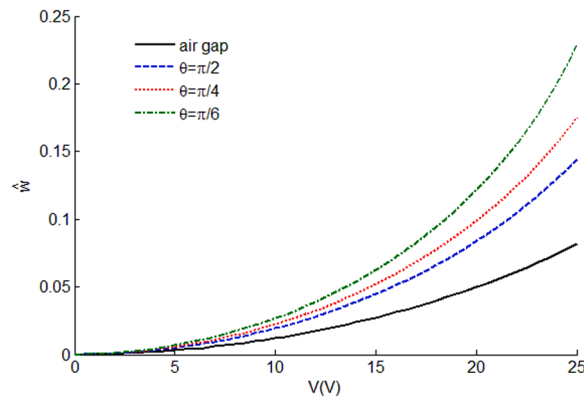


Fig. 4. variation of deflection of the beam's center point with the actuation DC voltage.

It is expected that the number of pillars can affect their deformation as well as the beam's deformation at a given actuation voltage. This effect is shown in Fig. 7, where the number of pillars is given by $N_p = \mu \frac{Lb(1 - \frac{L_p}{L} \cos(\theta))}{L_p b_p} = \mu \frac{bh}{A_p} (1 - \frac{L_p}{L} \cos(\theta))$, and μ is the ratio of the total surface area of pillars to the surface area of the beam. Reducing the number of pillars reduces the equivalent permittivity of the gap medium and therefore, as shown, reduces the deformation of the beam and pillars for a given voltage.

Fig. 8 displays the time response of the center point of the beam and central pillar endpoint to a 20 V step DC actuation voltage for various tilt angles and a PDMS viscosity of 0.005 N.s/m². The figure confirms that increasing the tilt angle results in a higher response amplitude of the system to a given stimulus. Furthermore, tilting the pillar results in an increase in transverse vibration amplitude, while the longitudinal amplitude decreases. This is due to the dominant compressive deformation of the pillars in a vertical state. Furthermore, at the vertical pillars where $\theta = \pi/2$, the transverse vibrations of the pillars will be zero due to the pure compressive deformation, as illustrated in part (b) of this figure.

Fig. 9 illustrates how the magnitude of the step DC voltage affects the response of the system. Parts (a) and (b) of the figure show the time histories for the beam's center point and the end point of the central pillar, respectively, while parts (c) and (d) correspond to their phase portraits. The amplitude of the longitudinal vibrations is smaller than that of the transverse vibrations of the pillar, which is related to the tilt angle of $\pi/6$. Fig. 8 confirms that when $\theta = \pi/4$, both amplitudes are approximately equal. However, for $\theta > \pi/4$, the longitudinal amplitude is greater than the transverse amplitude, and for $\theta < \pi/4$, it is the opposite. The decrease in vibration amplitude in Figs. 8 and 9 is due to the viscoelastic behavior of the pillars, which affects the beam's vibrations through the force applied at the connection points. Actually, the damping effects of the viscoelastic behavior depends on the value of the relaxation time in Eq. (1).

Capacitive MEMS devices, such as resonators, accelerometers, and densitometers, can be operated using an AC actuation voltage. However, these systems require a bias DC voltage to be applied initially, which can be used to tune the resonance frequency of the system. Fig. 10 illustrates the vibrational responses of the moving electrode and the central pillar through time responses and phase portraits, for 5 V DC+1 V AC amplitude with a dimensionless frequency of $\hat{\omega} = 10$. As demonstrated, there is a correlation between the number of pillars and the amplitude of the response. This is because the effective permittivity of the gap medium is higher for greater number of pillars.

Fig. 11 shows the system's time histories for three excitation frequencies when the bias actuation voltage is 10 V, in addition to an AC voltage with 5 V amplitude. The pillars' tilt angle is $\theta = \pi/6$, and μ is 0.3. Part (a) illustrates the behavior of the beam's center point, while parts (b) and (c) relate to the end point of the central pillar's transverse and longitudinal vibrations, respectively. At the resonance frequency, the system's vibration amplitudes are greater than those at lower or higher frequencies. The vibrations are dampened due to the viscoelastic behavior of PDMS, as well as the effective damping caused by the squeeze film or structural damping of the system.

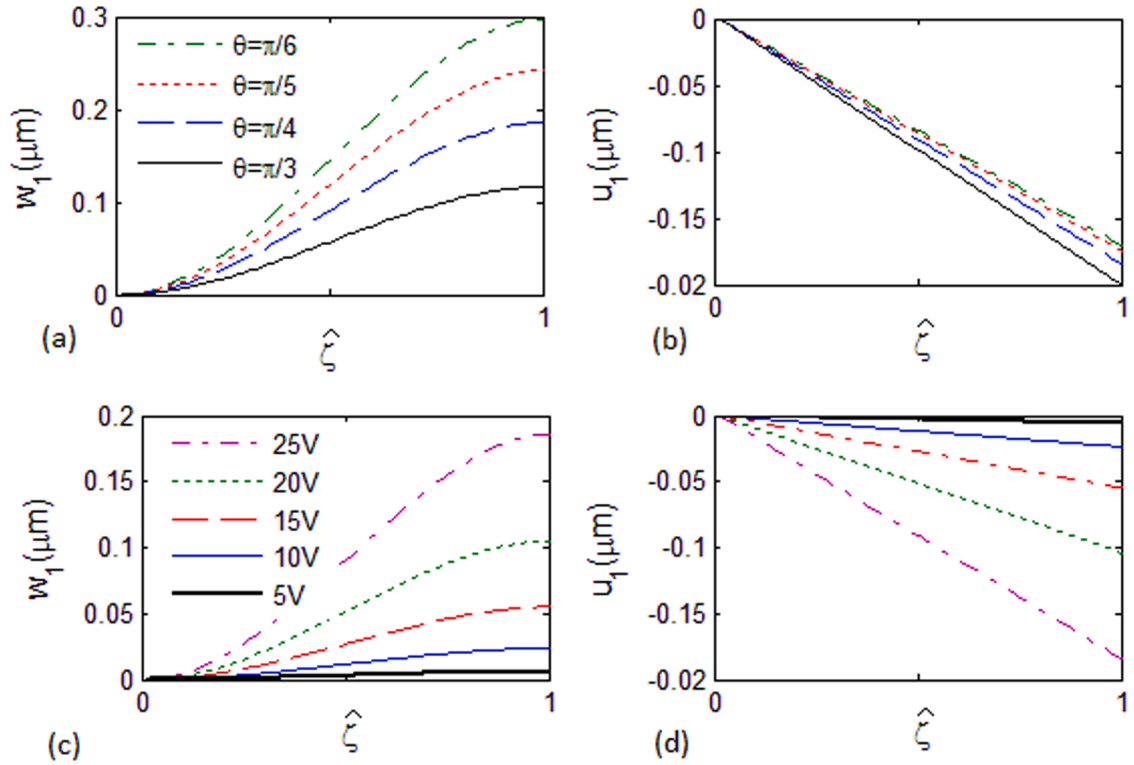


Fig. 5. Transvers and longitudinal deformations of the central pillar for: a and b) $V = 25$ V and different tilt angles, and c and d) $\theta = \frac{\pi}{4}$ and different voltages.

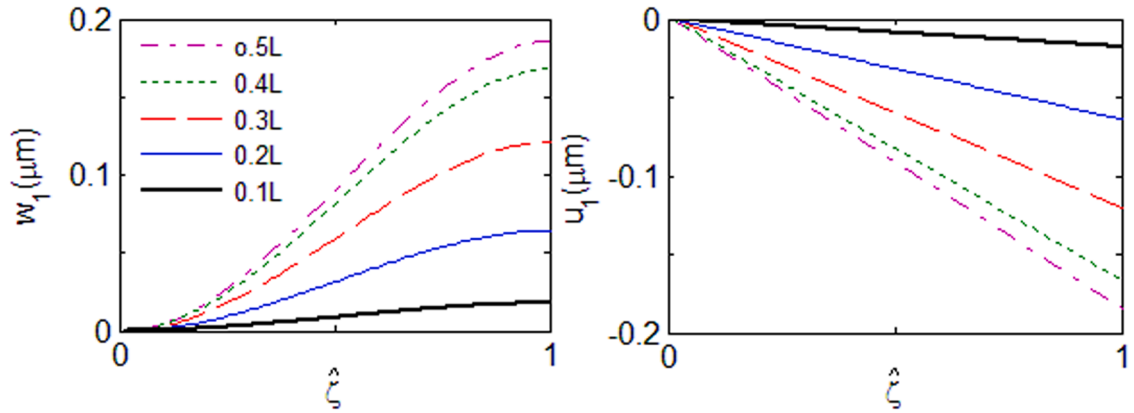


Fig. 6. Transverse and longitudinal deformations of the pillars located at different positions of connection to the beam for $V = 25$ V and $\theta = 45^\circ$.

5. Conclusion

This study presents a comprehensive mathematical model for an electrostatic MEMS with tilted micro-pillars. The model takes into account the viscoelastic properties of the PDMS pillars and incorporates coupled sets of three nonlinear differential equations that govern the longitudinal and transverse vibrations of the pillars, as well as the bending vibrations of the moving electrode. The results obtained from the model reveal important insights into the behavior of the device. The static deformations of the longitudinal and transverse directions of the pillars, as well as the bending deformation of the moving beam, were analyzed. It was observed that increasing the tilt angle leads to an increase in the longitudinal deformation while decreasing the other two deformations. This was due to the dominance of the compressive deformation of the pillars compared to their bending deformation at the tilt angles higher than $\theta = \pi/4$. This finding highlights the importance of considering the tilt angle in the design and optimization of such systems.

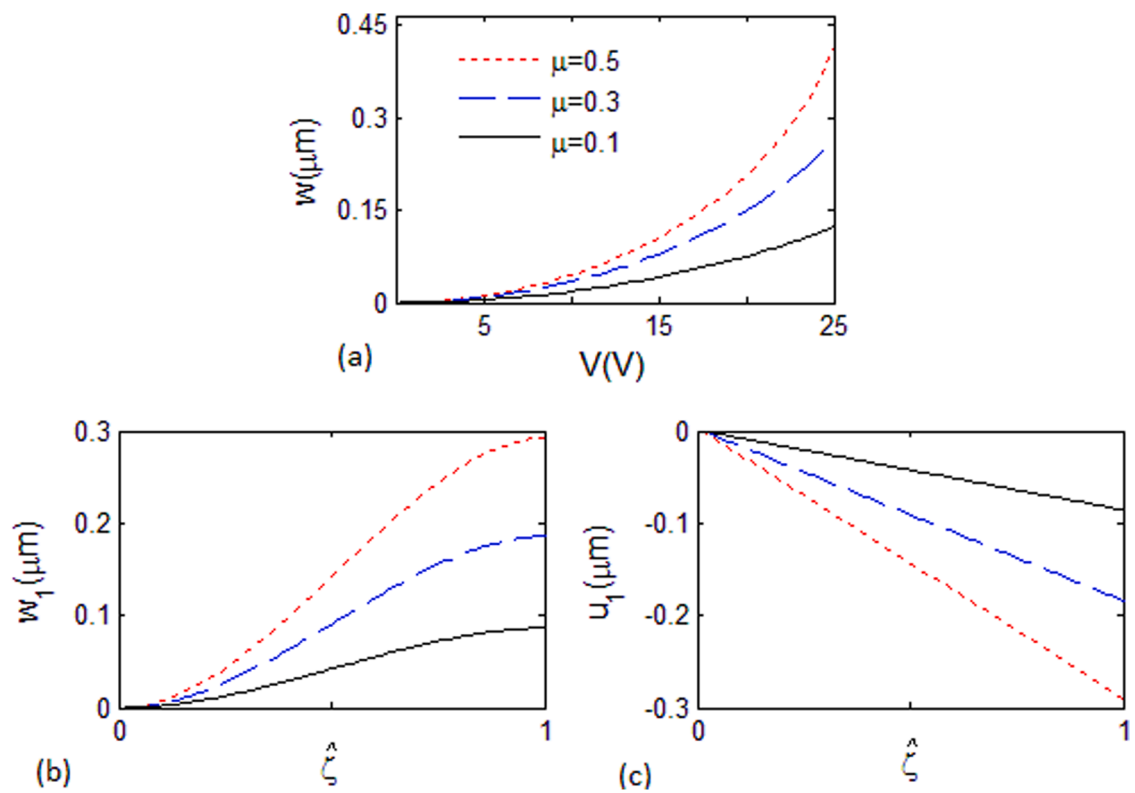


Fig. 7. Effect of the number of pillars on the: a) deflection of the center point of the beam versus voltage, b and c) pillars transverse and longitudinal deformation, at 25 V actuation voltage and $\theta = \frac{\pi}{4}$.

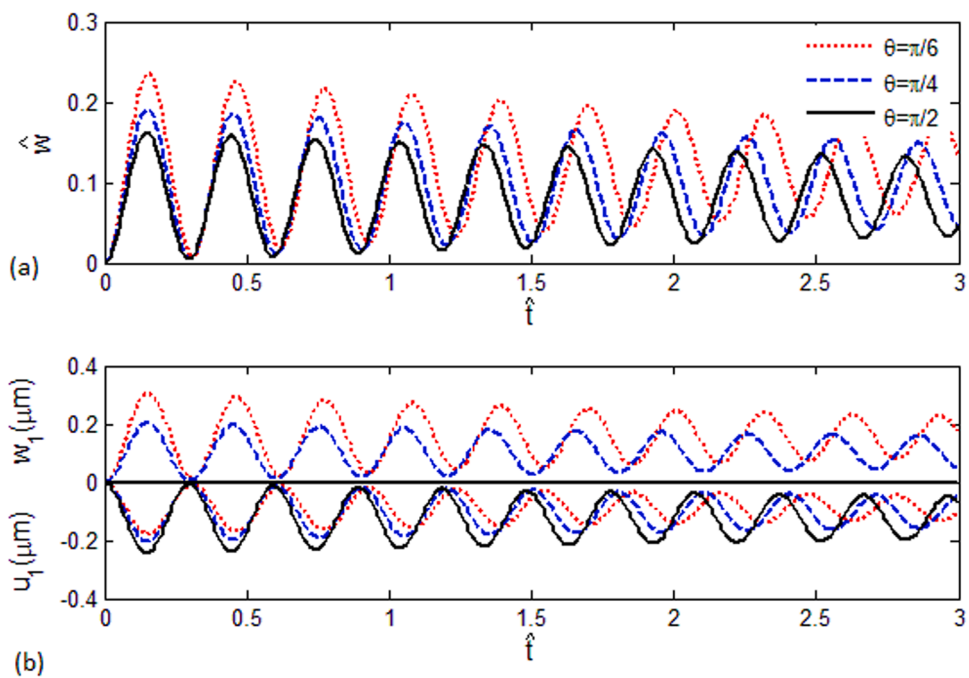


Fig. 8. Time history of a) transverse vibrations of the center point of the beam, and b) transverse and longitudinal vibrations of the end point of the central pillar.

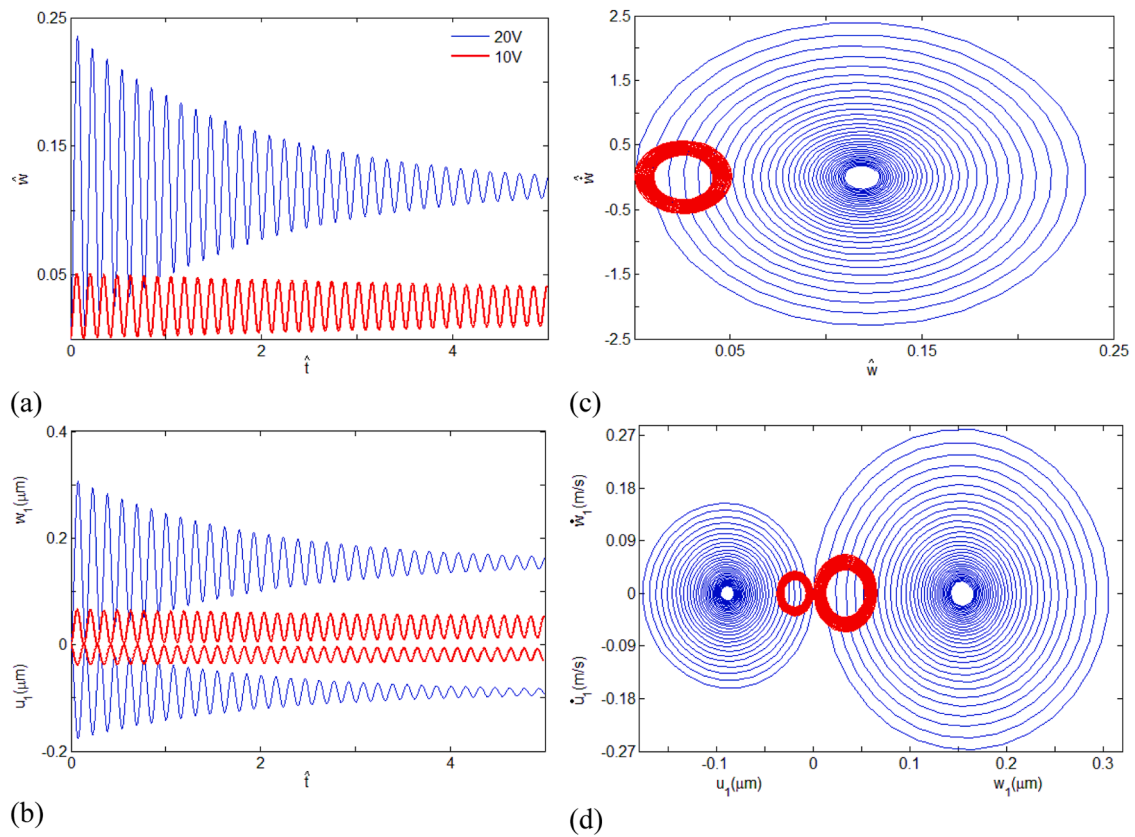


Fig. 9. Effect of the step DC actuation voltage on the: a, b) time response and c, d) phase portrait of the components of the system.

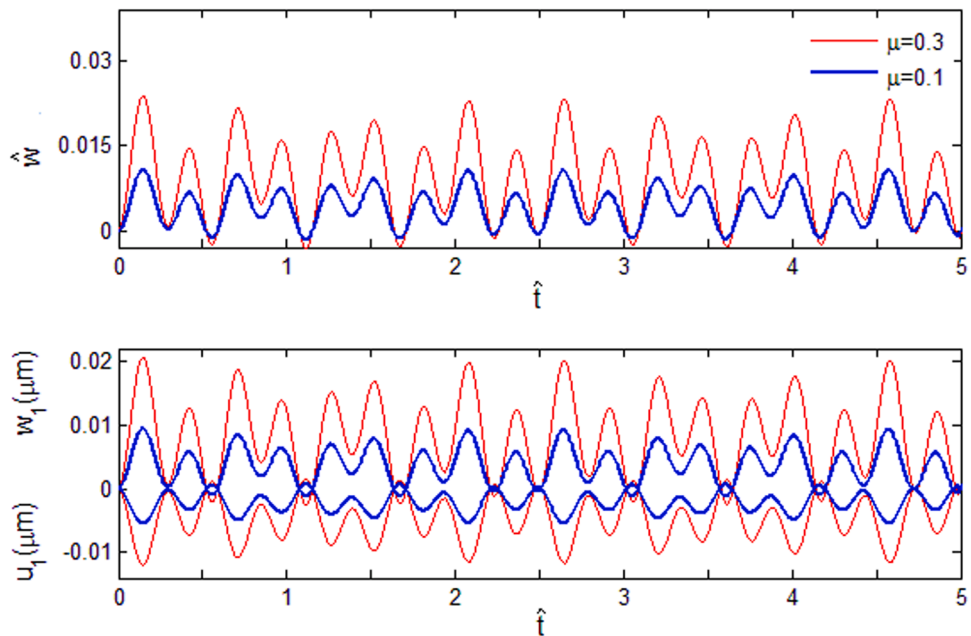


Fig. 10. Time histories of the a) center point of the beam and b) transverse and longitudinal vibrations of the central pillar with different μ .

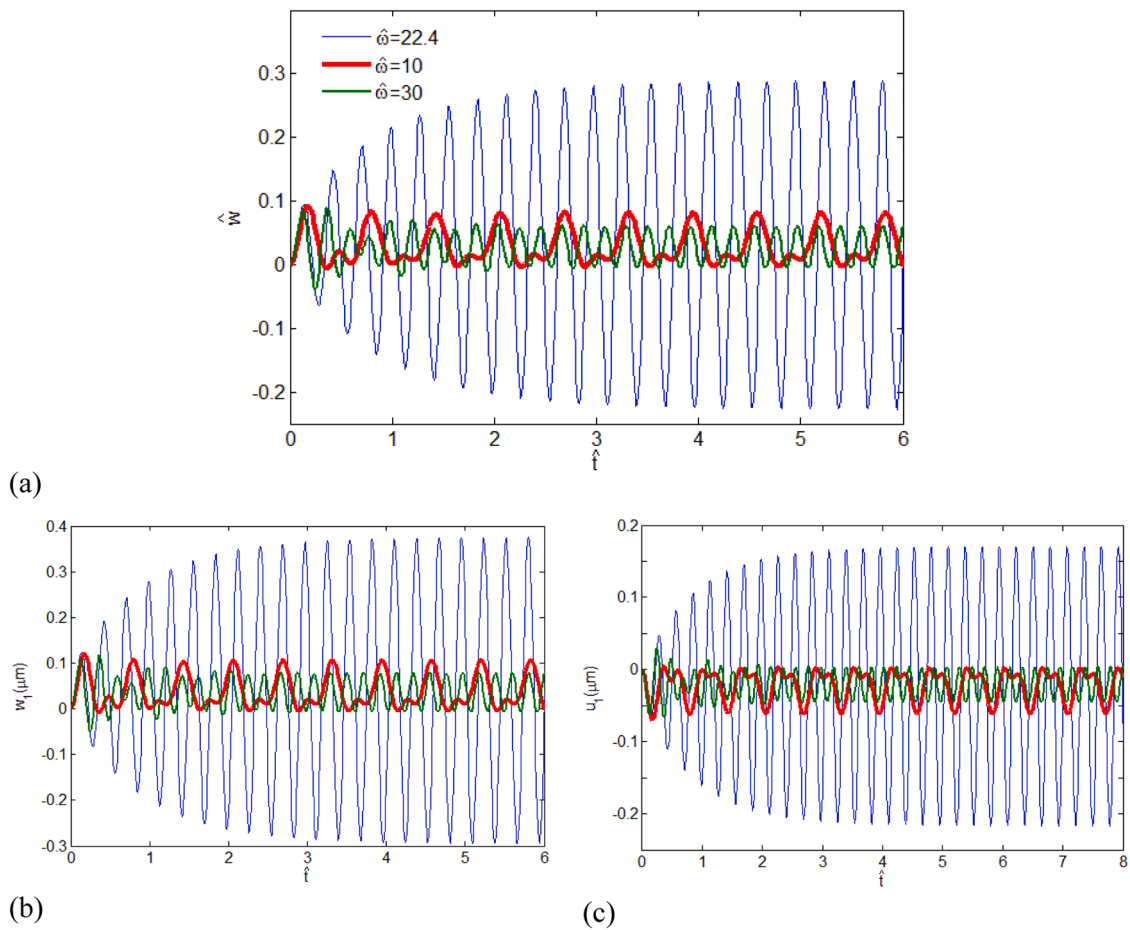


Fig. 11. Effects of the excitation frequency on the time responses of the a) center point of the beam, b) transverse, and c) longitudinal vibrations of the endpoint of the central pillar.

Additionally, it was found that increasing the number of pillars results in increased deformations due to the increased equivalent permittivity of the microstructured gap compared to an air gap. This demonstrates the influence of the pillar array on the device's behavior and suggests potential strategies for enhancing its performance.

Furthermore, the dynamic responses of the system under different actuations were investigated. The time histories and phase portraits of the system elements under step DC and DC+AC actuations were analyzed. The obtained results further validate the proposed model and its ability to accurately predict the behavior of capacitive MEMS devices with gap-filling arrays of tilted micro-pillars.

Overall, the comprehensive nonlinear mathematical model developed in this study serves as a valuable tool for accurately predicting the behavior of such devices. It enables a thorough understanding of their static deformations and dynamic responses under various actuation scenarios. The insights gained from this model can aid in the design, optimization, and performance evaluation of capacitive MEMS devices that utilize microstructured gaps with tilted micro-pillar arrays.

CRediT authorship contribution statement

Ahmed Hashim Kareem: Formal analysis, Investigation, Software. **Mohammad Fathalilou:** Formal analysis, Investigation, Methodology, Software, Supervision, Validation, Writing – original draft, Writing – review & editing. **Ghader Rezazadeh:** Validation, Writing – original draft, Conceptualization, Formal analysis, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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